Wavelet Analysis of Groundwater Level Fluctuations in Long Island and the Metropolitan Area, NY

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Abstract:

Long Island's groundwater levels fluctuate annually over time due to precipitation, human activities, the aquifer's ability to replenish, and other possible geological and atmospheric factors. This research investigates the relationship between groundwater periodicity and human interference to visualize the probable effects on the water supply throughout Long Island between the years 2000 to 2025. This study includes the possible impacts of human interference across Long Island's groundwater system using wavelet analysis to compare the effects caused before, during, and after the COVID-19 pandemic in 2020. The methods within this research incorporate coding of groundwater data in active groundwater stations reported by the monitoring wells of the USGS (United States Geological Survey) database using R programming language in RStudio. Stations are located within large industries like shopping areas, laboratories, universities, and even in parks throughout NYC, White Plains, Nassau, and Suffolk County.

Wavelet analysis figures were constructed containing the groundwater data that was utilized to visually determine the quantity of groundwater before, during, and after the pandemic period based on the patterns of strong and weak fluctuations within the groundwater system. The separation of long-term and short-term data from the raw wavelet figures aided in uncovering oscillating patterns. This oscillation frequency indicates a re-charging or discharging component in the aquifer. The raw and long-term wavelets of groundwater levels reveal stronger frequencies sometime before the pandemic, weak to no frequencies during the pandemic, and some indications of stronger frequencies returning after the pandemic. However, short-term wavelets of groundwater levels appeared to show random patterns that weren't as abundant as the raw and long-term wavelets. Precipitation data patterns seem relatively consistent except for a slight decrease in the overall precipitation pattern shortly after 2015, indicating another factor, other than human interference during COVID-19, potentially causing this phenomenon. Therefore, the weaker fluctuations in groundwater levels during the year 2020 are possibly due to the COVID-19 pandemic, when large industries went out of business and stopped using groundwater for their large productions. The use of wavelet analysis allows for a more visual perceptive to display how the frequencies change over time. This technique can be used to monitor and ensure the stability of freshwater accessible to Long Island.

Background of Study:

- Analyzes groundwater level and precipitation patterns based on COVID-19 impacts
- Long Island's groundwater relies solely on precipitation,
- Wavelet Analysis Approach: Detects periodic patterns and disruptions in groundwater levels to assess human impact on natural recharge cycles.
- Groundwater vs. precipitation wavelets—disruptions in expected patterns may indicate human influence.
- Findings will help understand groundwater system responses to changing water use patterns and inform future resource management.

Precipitation Wavelets:

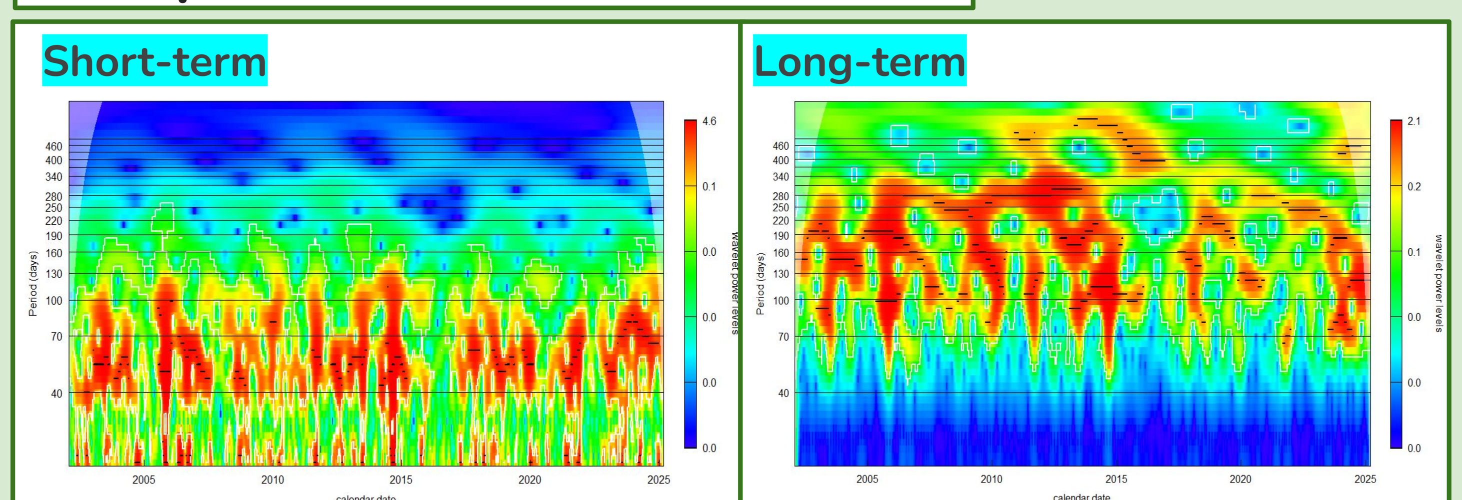
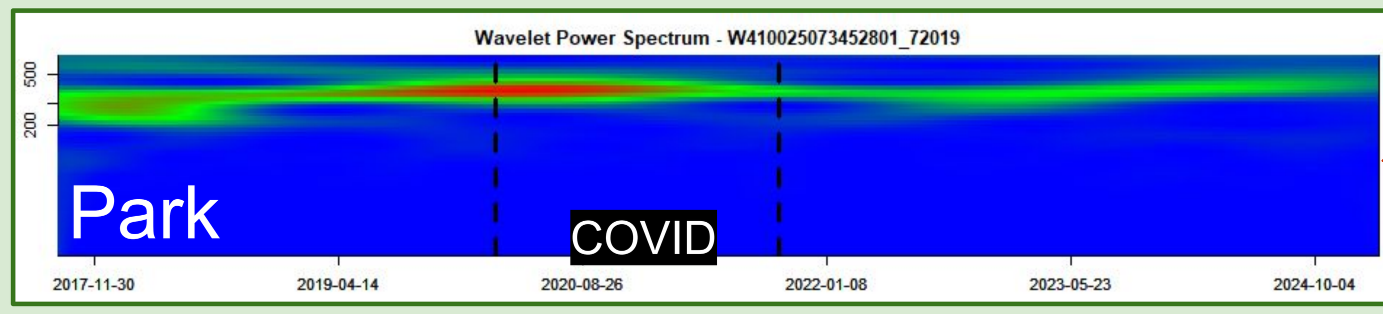
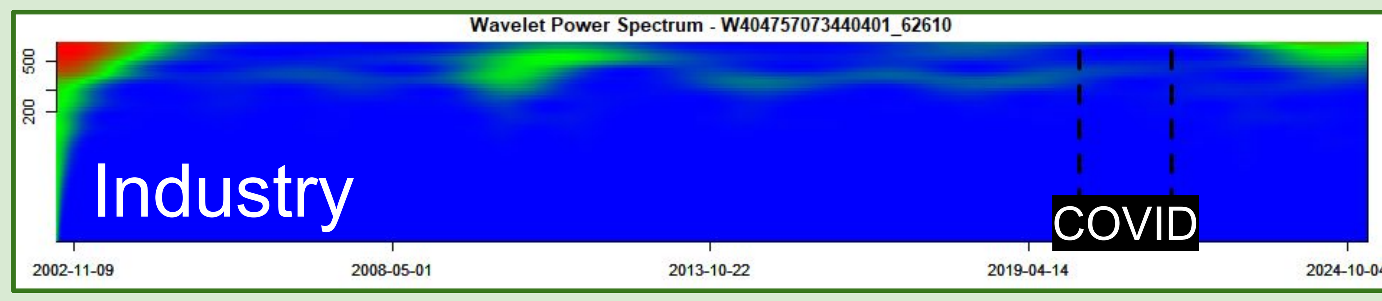


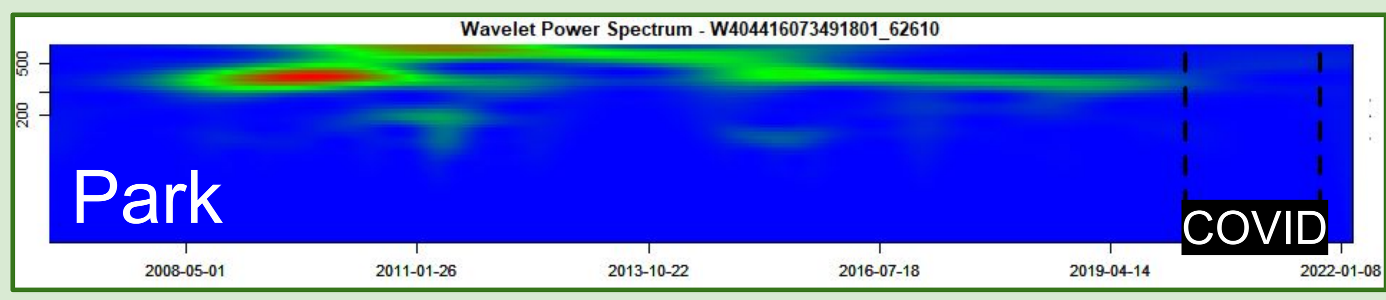
Figure 2: The short and long-term wavelet analysis of precipitation data from the Islip-Macarthur Airport.

Method/Materials:

- Precipitation data used to create wavelets is collected from NOAA (Fig. 2)
- Daily groundwater data collected from the USGS monitoring well database
  - 58 total stations
- Filtered all data stations between Feb 1, 2000 - Feb 10, 2025
- **RStudio (v.4.2.1)**
  - Interpolated gaps (maximum 14 day consecutively) within the data using na.approx ( ) function
  - Plotted groundwater data into a times series to detect interpolated gaps
  - Plotted coordinates for **10** useable groundwater stations onto a map for spatial analysis (Fig.1)
  - Performed Wavelet Comp package in R utilizing "wavelet.analysis" function
  - Subtracted moving average from raw data to display the long-term and short-term frequencies







Results:

- **Study Area**: Data from 10 groundwater monitoring stations across Long Island and the metropolitan area; precipitation data from Islip-Macarthur Airport.
- **Precipitation Findings**: No major disruptions in precipitation fluctuations during the 2020 pandemic, but a slight decrease after 2015. (Fig. 2)
- **Groundwater Findings**: Most stations showed strong fluctuations before 2020, weak or no fluctuations during 2020, and weak fluctuations resuming after 2020. (Fig. 1)
- **Exceptions**: Two stations (within parks) showed strong fluctuations throughout, while stations near industries showed varying patterns. (Fig. 1)
- **Short-Term vs. Long-Term Trends**: Separation of short-term and long-term data revealed clearer distinctions in fluctuation patterns.
- **Short-Term Groundwater Data**: Appeared random and inconsistent with raw and long-term groundwater trends.

Discussion/Conclusion:

- **Groundwater is the sole source of drinking water for L.I.**
- **L.I. aquifers' only inflow is precipitation**
  - **Runoff captured by recharge basins**
- **Rainwater data collected by Islip-MacArthur Airport Precipitation Station (Precipitation)**
- **Only change in pattern came in 2015**
  - **Likely ENSO phenomena**
- **No change during COVID-19**
- **However...**
  - **One station**
  - **No snow data**
- **Human intervention to groundwater level**
- **Isolating COVID-19 gives exaggerated results**
  - **Lockdown**
- **Used long-term data wavelets**
  - **Pandemic affects likely a few years**

What are we looking for?

- **Industrial /commercial areas to have strong usage before lockdown, weak after**
- **Park areas to not be affected**

Suffolk

- **Industrial/commercial stations + beach matched pattern (3 stations)**
  - **Strong oscillations before, weak-no oscillations during, weak oscillations after**
- **Park showed different pattern (1 station)**
  - **Strong, continuous oscillations throughout**
- **Matches hypothesis**

Nassau

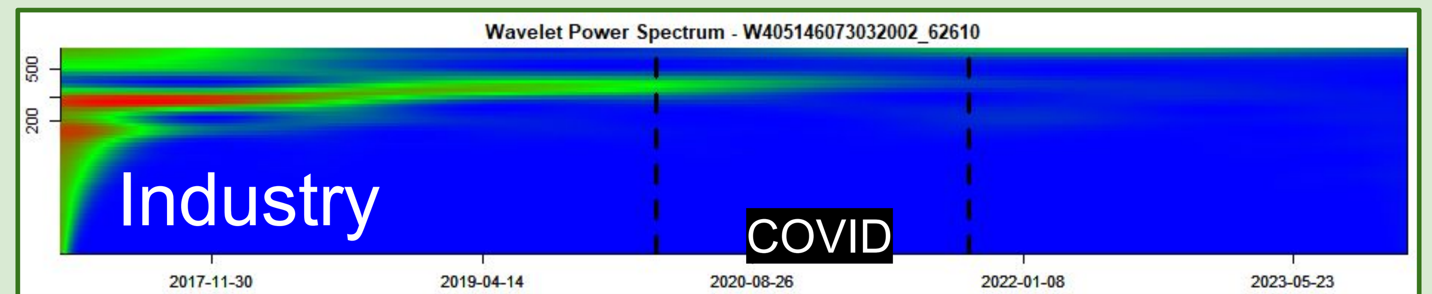
- **Industrial/commercial stations matched pattern (3 stations)**
  - **Strong oscillations before, weak-no oscillations during, weak oscillations after**
- **Matches hypothesis**
- **Corona Park, Kissena Golf Course, parks north of Manhattan**

Queens

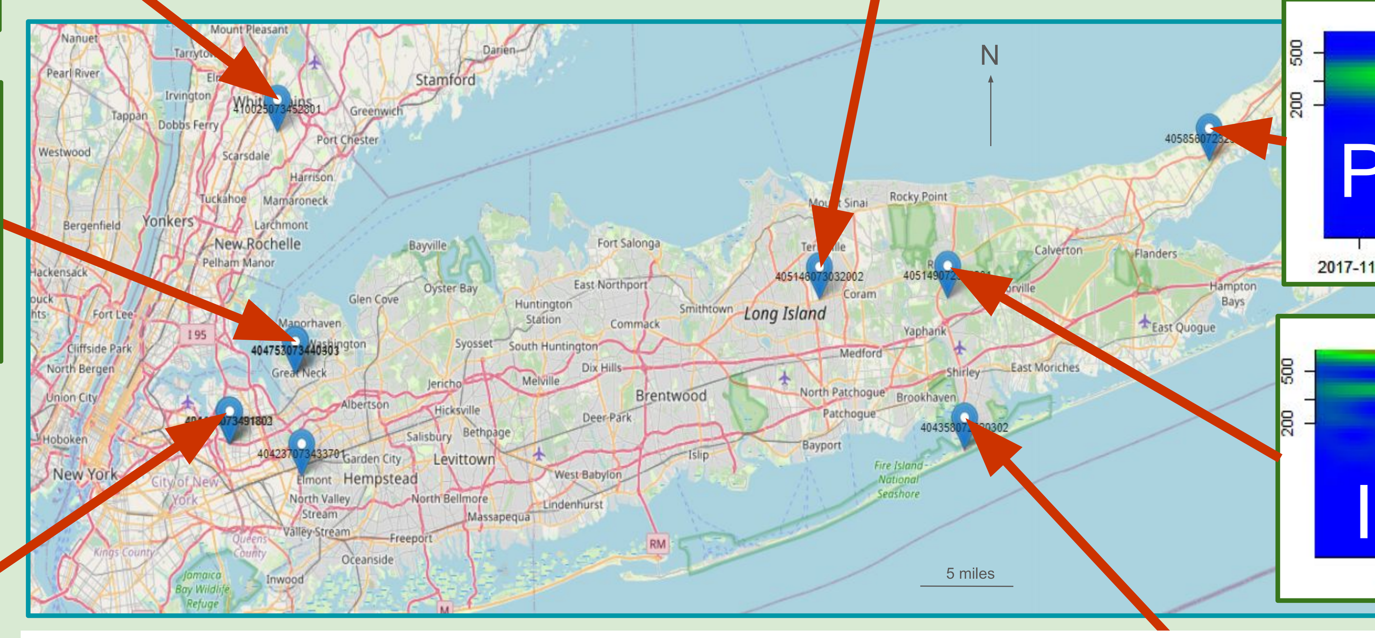
- **Parks (2 stations)**
  - **Strong oscillations before, weak-no oscillations during, weak oscillations after**
- **Didn't match hypothesis**

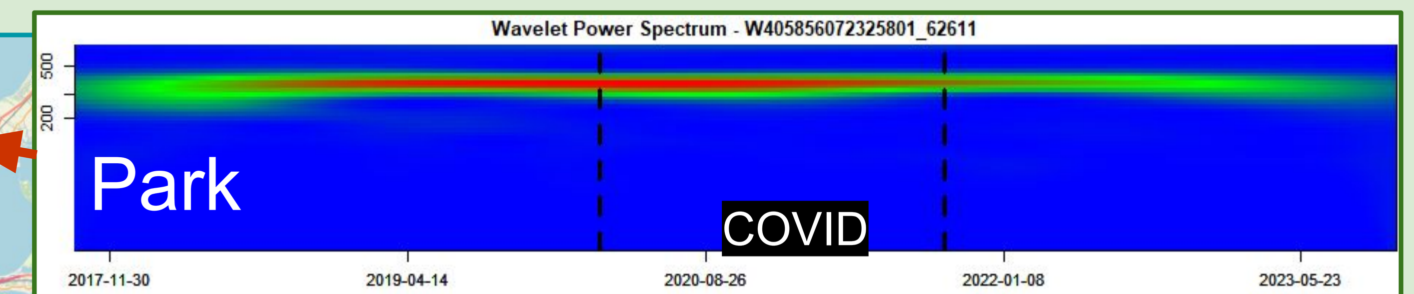
White Plains

- **Park matched pattern (1 station)**
  - **Strong, continuous oscillations throughout**
- **Matched hypothesis**



Hypothesis: Industrial areas, like shopping areas and universities, show weak to no groundwater oscillations during 2020. Parks in Queens would show no changes in groundwater oscillations due to groundwater taken from Upstate New York





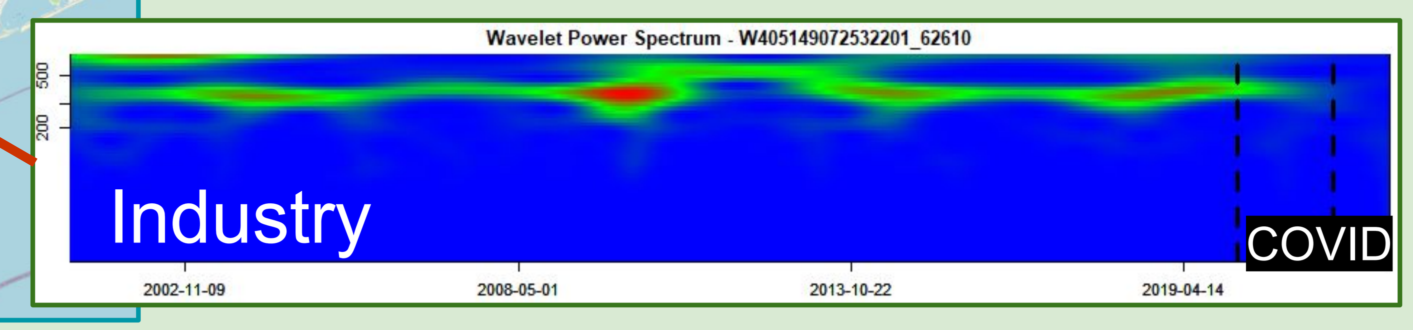
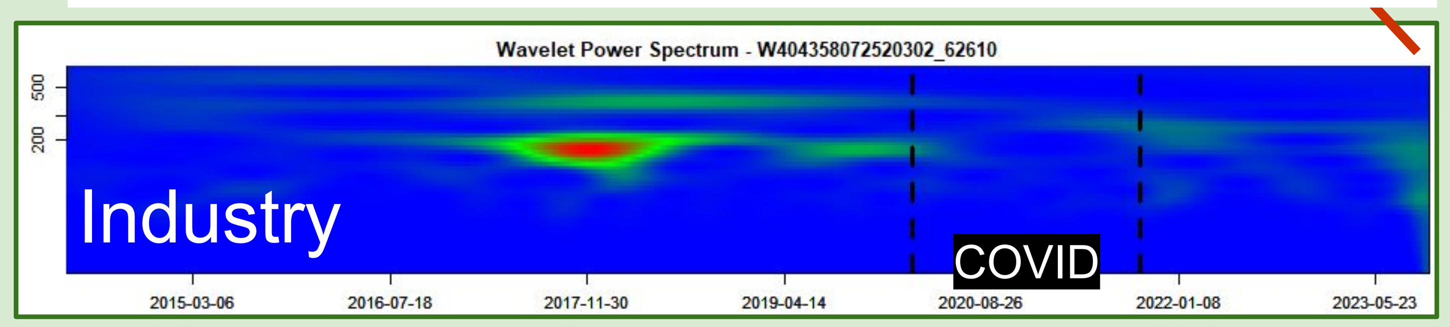


Figure 1: Map displays the 10 groundwater monitoring well stations, located at the east and the west of Long Island, studied in this research. Wavelets of the long term data (~200-500 days, y axis) with red and green areas showing stronger fluctuations in groundwater levels, blue showing weaker. Covid dates shown as black dashed lines.



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Credit Authorship Contribution Statement:

Parag, A: Literature, Editing, RStudio Coding, Writing- Abstract, Introduction, Methods, Results, Results- Figures 1-5. Jones, W: Literature, Editing, RStudio Coding, Writing- Methods, Discussion, Results- Figure 2. Cann, I: Literature, Editing, RStudio Coding, Results- Figures 6,7,8. Bradley, C: Literature, Editing, RStudio Coding, Writing- Conclusion

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